

Hands-on Activity 4.4 Sass Essentials Lab Project	
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1. Discussion	
This module tackles the use of SASS or Syntactically Awesome Style Sheets. CSS preprocessors like Sass and Less were created to solve the limitations of traditional CSS, which can become repetitive and hard to manage in large projects. Instead of writing plain .css files directly, developers write styles in .scss or .less files, which are then compiled into standard CSS. Between the two, Sass and Less are similar but differ slightly in syntax and popularity. For example, Sass uses the \$ symbol for variables, while Less uses @. Although both are powerful, Sass is currently the more widely adopted industry standard. Their core features include variables and nesting. Variables let developers define reusable values once and apply them throughout the stylesheet, improving consistency and maintainability. Nesting allows selectors to be structured hierarchically to reflect HTML structure, making styles more readable and organized.	
3. Procedure	
Scenario In this lab, you will learn the basics of Sass (Syntactically Awesome Style Sheets) and apply them to enhance a simple web page. You will practice using variables, mixins, and nesting to make your CSS more efficient and maintainable. You can use an online Sass compiler like SassMeister or the Sass Playground to write and compile your Sass code. Instructions Follow the instructions below to create the webpage step by step: 1. Setting Up Sass Open an online Sass compiler like the Sass Playground (external link). 2. Creating Variables Define variables for the primary color and font size in your Sass code. Use these variables to style the body text and background color of the header and footer. Create a variable named \$primary-color with the value #007bff and a variable named \$font-size with the value 16px. Apply the \$primary-color variable to the color property of the body element. Apply the \$font-size variable to the font-size property of the body element. Set the background-color of the .header and .footer to a lightened version of the \$primary-color using the lighten function. Add padding of 20px to .header and .footer, and center-align their text. Show Sample Implementation 3. Using Mixins Create a mixin named rounded-corners to add rounded corners to elements.	

The mixin should accept a parameter `$radius` and apply it to the `border-radius` property. Use this mixin to style the `li` elements inside `.feature-list` and the `.example-text` element. Set `border-radius` to `10px` using the mixin. Set the padding to `10px` and background-color to a lightened version of the `$primary-color` for both `.feature-list li` and `.example-text`.
Show Sample Implementation

4. Nesting

Nest the styles for the elements inside the `.header`, `.about`, `.features`, and `.example` sections to reflect their structure in the HTML.

In the `.header`, nest an `h1` element and set its font-size to `2em`. In the `.about`, nest an `h2` element and set its font-size to `1.5em`, and nest a `p` element and set its font-size to the `$font-size` variable. In the `.features`, nest an `h2` element and set its font-size to `1.5em`. Then nest `.feature-list` and apply `list-style-type: disc` and `padding-left: 20px`. Finally, nest `li` elements inside `.feature-list` and use the `rounded-corners` mixin with a `5px` radius, add `padding: 5px`, `margin-bottom: 5px`, and background-color as a lightened version of the `$primary-color`. In the `.example`, nest an `h2` element and set its font-size to `1.5em`, and nest `.example-text` and use the `rounded-corners` mixin with a `10px` radius, add `padding: 10px`, and background-color as a lightened version of the `$primary-color`.
Show Sample Implementation

5. Compiling and Testing

Copy your Sass code into the online compiler and compile it to CSS. Replace the content of `styles.css` in your project with the compiled CSS.

Download your files and open `index.html` in a local browser to see the styled page. Make any necessary adjustments to your Sass code.

4. Output

HTML CODE

```
HTML
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Sass Essentials</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header class="header">
    <h1>Welcome to Sass Essentials</h1>
  </header>
  <main>
    <section class="about">
```

```

        <h2>About Sass</h2>
        <p>Sass makes CSS more powerful with features like
variables, mixins, and nesting.</p>
    </section>
    <section class="features">
        <h2>Features</h2>
        <ul class="feature-list">
            <li>Variables</li>
            <li>Mixins</li>
            <li>Nesting</li>
        </ul>
    </section>
    <section class="example">
        <h2>Example</h2>
        <p class="example-text">This text will be styled using
Sass.</p>
    </section>
</main>
<footer class="footer">
    <p>&copy; 2035 Sass Essentials</p>
</footer>
</body>
</html>

```

OUTPUT

Welcome to Sass Essentials

About Sass

Sass makes CSS more powerful with features like variables, mixins, and nesting.

Features

- Variables
- Mixins
- Nesting

Example

This text will be styled using Sass.

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5. Supplementary Activity

CODE

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Cat World</title>
  <link
href="https://fonts.googleapis.com/css2?family=Caveat:wght@400;700&family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="supple.css">
</head>
<body>

<section class="hero">
  <div class="hero-content">
    <h1>Welcome to Cat World 🐾</h1>
    <p>Discover adorable and amazing cat breeds</p>
    <button>Explore Cats</button>
  </div>
</section>

<section class="about-section">

  <div class="about-container">
    <div class="about-text">
      <h2>About Cats 🐾</h2>
      <p>
        Cats (Felis catus) are small carnivorous mammals that
        have been
        domesticated for thousands of years. There are over 70
        recognized
        cat breeds worldwide, each with unique characteristics,
        coat types,
        and personalities.
      </p>

      <p>
        With cats, always be ready for a fun and crazy
        adventure. They can take
        you for a crazy wild cat ride.
      </p>
    </div>

    <div class="video-card">
      <video controls>
        <source src="posaaa.mp4" type="video/mp4">
        Your browser does not support the video tag.
      </video>
    </div>
  </div>
</section>
```

```
<section class="cards-section">
  <h2 class="section-title">Popular Cats</h2>
  <div class="card-container">
    <div class="card">
      
      <div class="card-content">
        <h3>Persian Cat</h3>
        <p>Known for their long fluffy coat and calm
personality.</p>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Siamese Cat</h3>
        <p>Elegant and vocal cats with striking blue eyes.</p>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Maine Coon</h3>
        <p>One of the largest domestic breeds</p>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Bengal Cat</h3>
        <p>Energetic and exotic-looking with a leopard-like coat
pattern.</p>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Ragdoll</h3>
        <p>Gentle and affectionate cats known for their striking
blue eyes.</p>
      </div>
    </div>
  </div>
</section>
```

```
</body>
</html>
```

SCSS

CSS

```
$primary-orange: #FF8C42;
$soft-orange: #FFA94D;
$dark-orange: #CC6B2C;
$light-bg: #FFF3E6;
$about-bg: #FFE8D6;
$text-dark: #333;
$text-muted: #555;

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Poppins', sans-serif;
  background-color: $light-bg;
}

.hero {
  height: 100vh;
  background: linear-gradient(
    rgba(255, 140, 66, 0.8),
    rgba(255, 140, 66, 0.8)
  ),
  url('https://images.unsplash.com/photo-1518791841217-8f162f1e1131');

  background-size: cover;
  background-position: center;
  display: flex;
  justify-content: center;
  align-items: center;
  text-align: center;
  color: white;

  h1 {
```

```
    font-family: 'Caveat', cursive;
    font-size: 60px;
  }

  p {
    margin: 15px 0;
    font-size: 20px;
  }

  button {
    padding: 12px 25px;
    border: none;
    border-radius: 8px;
    background-color: $soft-orange;
    font-size: 18px;
    cursor: pointer;
    transition: 0.3s;

    &:hover {
      background-color: $dark-orange;
      color: white;
    }
  }
}

.about-section {
  padding: 100px 20px;
  background-color: $about-bg;

  .about-container {
    max-width: 1200px;
    margin: 0 auto;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 80px;
    flex-wrap: wrap;
  }

  .about-text {
    flex: 1;
    min-width: 300px;
    max-width: 600px;

    h2 {
      font-family: 'Caveat', cursive;
      font-size: 45px;
      color: $dark-orange;
      margin-bottom: 20px;
    }
  }
}
```

```

    }

    p {
      font-size: 16px;
      line-height: 1.6;
      margin-bottom: 15px;
      color: #444;
    }
  }

  .about-video {
    flex: 1;
    display: flex;
    justify-content: center;
  }

  .video-card {
    width: 300px;
    aspect-ratio: 9 / 16;
    border-radius: 25px;
    overflow: hidden;
    background: white;
    border: 5px solid $soft-orange;
    box-shadow: 0 20px 40px rgba(0,0,0,0.15);
    transition: 0.3s ease;

    &:hover {
      transform: scale(1.05);
    }

    video {
      width: 100%;
      height: 100%;
      object-fit: cover;
    }
  }
}

.cards-section {
  padding: 100px 20px;
  text-align: center;

  .section-title {
    font-family: 'Caveat', cursive;
    font-size: 45px;
    margin-bottom: 50px;
    color: $dark-orange;
  }
}

```



```
.card-container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 30px;
  max-width: 1200px;
  margin: 0 auto;
}

.card {
  background: white;
  border-radius: 15px;
  overflow: hidden;
  box-shadow: 0 8px 20px rgba(0,0,0,0.1);
  transition: 0.3s;

  &:hover {
    transform: translateY(-10px);
  }

  img {
    width: 100%;
    height: 200px;
    object-fit: cover;
  }

  .card-content {
    padding: 20px;

    h3 {
      font-family: 'Caveat', cursive;
      font-size: 28px;
      margin-bottom: 10px;
      color: $primary-orange;
    }

    p {
      font-size: 14px;
      color: $text-muted;
    }
  }
}
}
```

OUTPUT

Welcome to Cat World 🐾🐾

Discover adorable and amazing cat breeds

Explore Cats

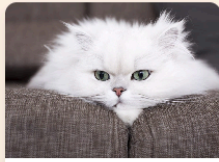
About Cats 🐾🐾

Cats (*Felis catus*) are small carnivorous mammals that have been domesticated for thousands of years. There are over 70 recognized cat breeds worldwide, each with unique characteristics, coat types, and personalities.

With cats, always be ready for a fun and crazy adventure. They can take you for a crazy wild cat ride.

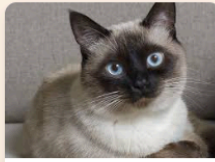


Popular Cats



Persian Cat

Known for their long fluffy coat and calm personality.



Siamese Cat

Elegant and vocal cats with striking blue eyes.



Maine Coon

One of the largest domestic breeds



Bengal Cat

Energetic and exotic-looking with a leopard-like coat pattern.



Ragdoll

Gentle and affectionate cats known for their striking blue eyes.

CONCLUSION

After performing this activity I was able to learn about the use of SASS in building websites. In learning the method of styling through SASS I learned that we can utilize the use of nesting and functions for repetitive code blocks so that we can reuse them making the code better. I was able to use SASS in a small sample animal website wherein I was able to use SASS for properly styling the website. For me I had a hard time following this style because it is new to me. There was also an issue for the setup but I resolved it by following the steps in the documentation. Overall, I observed the styling code was shorter than normal css but still able to style my website accordingly which helps in improving the work standards I can use in creating new websites.